Aiding Software Root Cause Analysis with Large Language Models

- Evaluation of the Effectiveness of Fine-tuned T5, GPT, and RAG in the handling Customer Fault Reports

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Abstract

Software systems generate a substantial number of fault reports during pre-deployment customer testing, making manual root cause analysis (RCA) both time-consuming and error-prone. This study explores the use of large language models (LLMs)—specifically T5, GPT-2, and a retrieval-augmented generation (RAG) model—to automate and enhance the RCA process in a domain-specific software engineering setting. Using a curated dataset of real-world fault descriptions and resolutions, the models were fine-tuned and evaluated using BLEU-4, ROUGE, and BERT-based semantic similarity metrics. Results indicate that T5 outperforms GPT-2 in lexical and structural fidelity (e.g., BLEU-4: 0.1810 vs. 0.1210), while RAG achieves the highest semantic similarity (BERT score: 0.7715). These findings suggest that combining T5's precision in technical phrasing with RAG's contextual understanding may offer a promising direction for developing intelligent RCA assistance tools that improve both accuracy and relevance in software fault diagnosis. Future work will focus on hybrid model optimization and user-centered system integration for real-world engineering workflows.

I. Introduction

Artificial intelligence has been increasingly applied in vehicle systems [1]. Recent research also highlights the role of reinforcement learning in fault detection [2].

Background

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- Second item
- Third item

- Forth item
- Fifth item